

Issues and Cybersecurity

YEAR 9 COMPUTING TECHNOLOGY

Learning Intention



Investigate issues with the use of data,
including cyber safety, security, privacy and ethics



Investigate data collection and interpretation adhering to privacy
and cybersecurity principles, including specify what data is collected, who owns it, and how it will be protected

Success Criteria



I can analyse issues with data use including cyber safety, security, privacy and ethics



I can explain how data is collected, who owns it, and how it will be protected

Information Systems – Considerations and Cybersecurity



Information systems operate within a **real-world context**



Successful systems consider:

People
Processes
Technology
Environment



Cybersecurity is essential for systems that operate online

Information System Context Diagrams



A context diagram shows:

- System components
- People and processes
- Data and information
- External environment



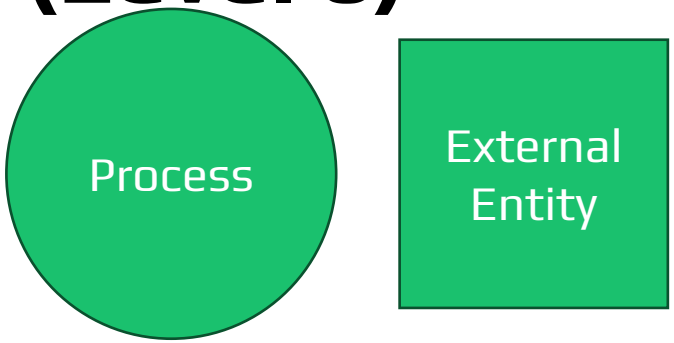
Helps identify:

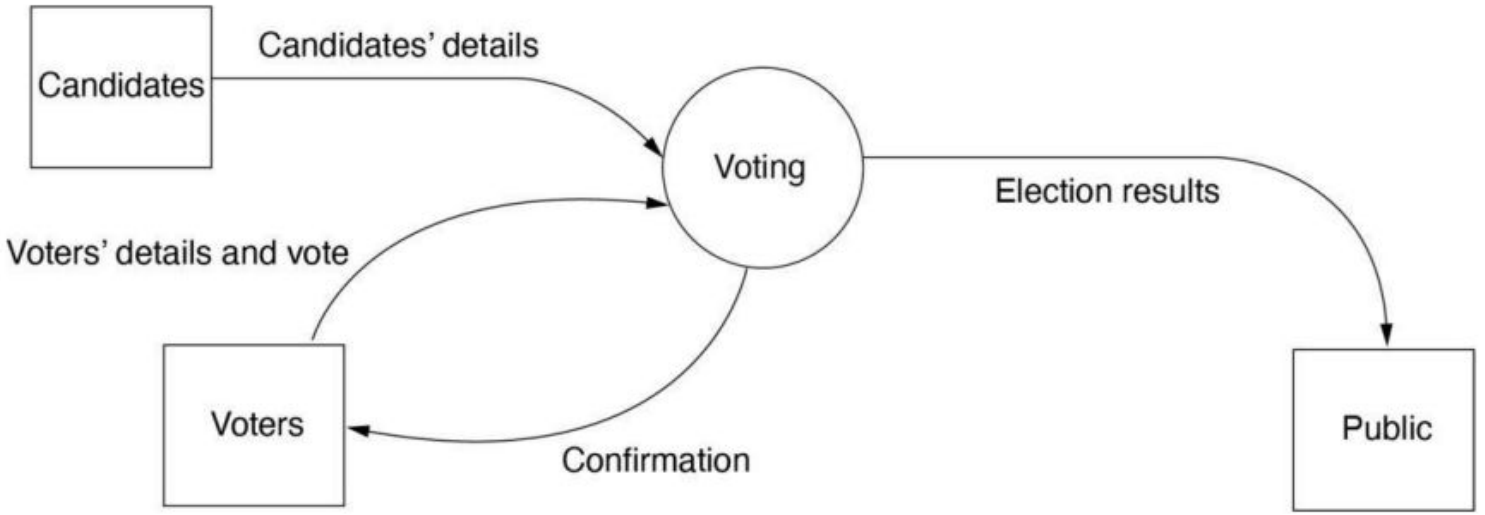
- What impacts the system
- What the system impacts



Used when designing or documenting systems

Data Flow Diagram / Context Diagram (Level 0)







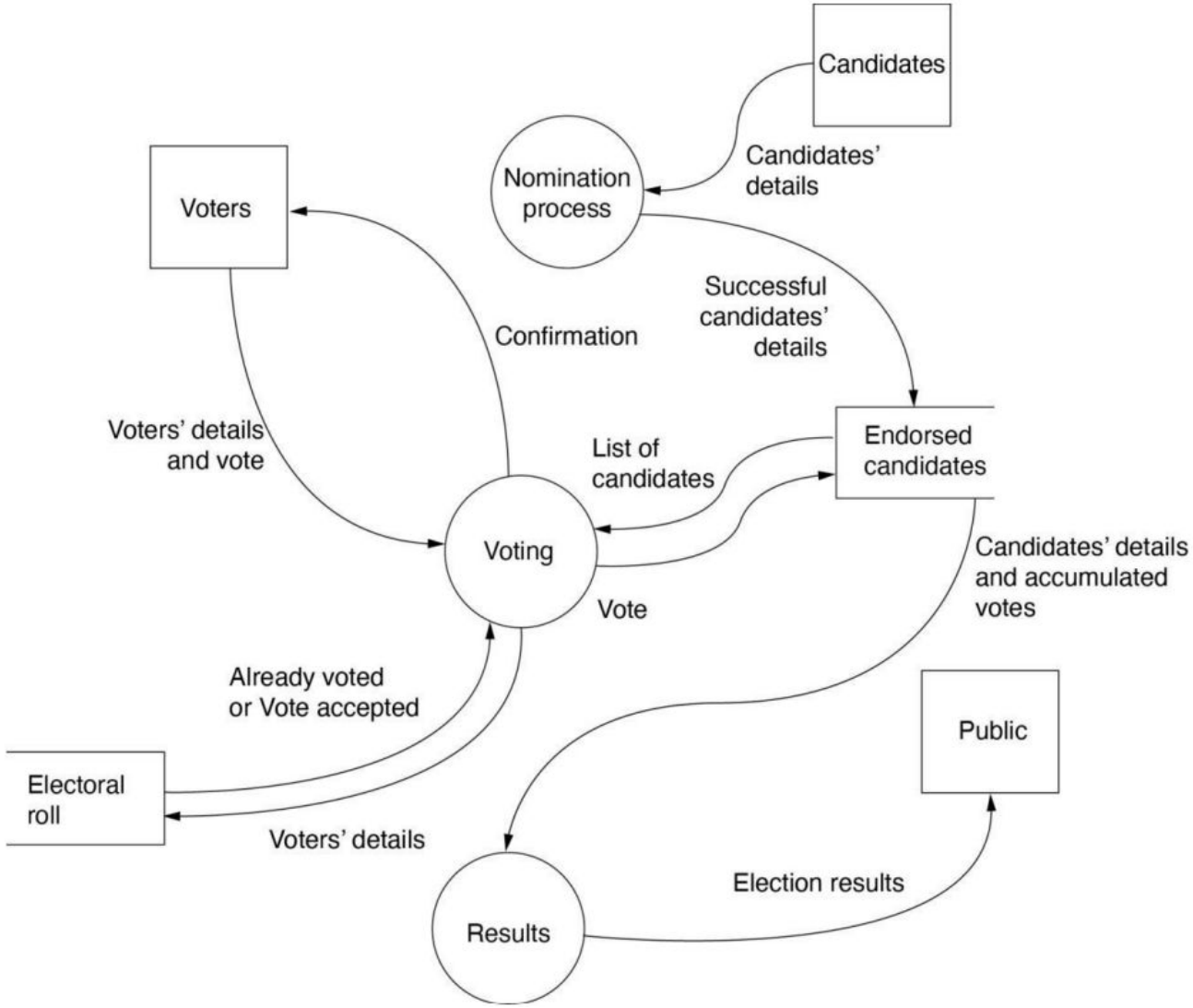
Data Flow Diagram (Level 1)

Data Store

Process

External Entity

Data Flow



Sociological Considerations



Systems can have **positive and negative social impacts**



Designers must consider:

- Privacy
- Equity
- Accessibility
- User trust



Example:

- Opal card improves convenience
- Anonymous cards reduce privacy concerns

Inclusive and Ethical Design

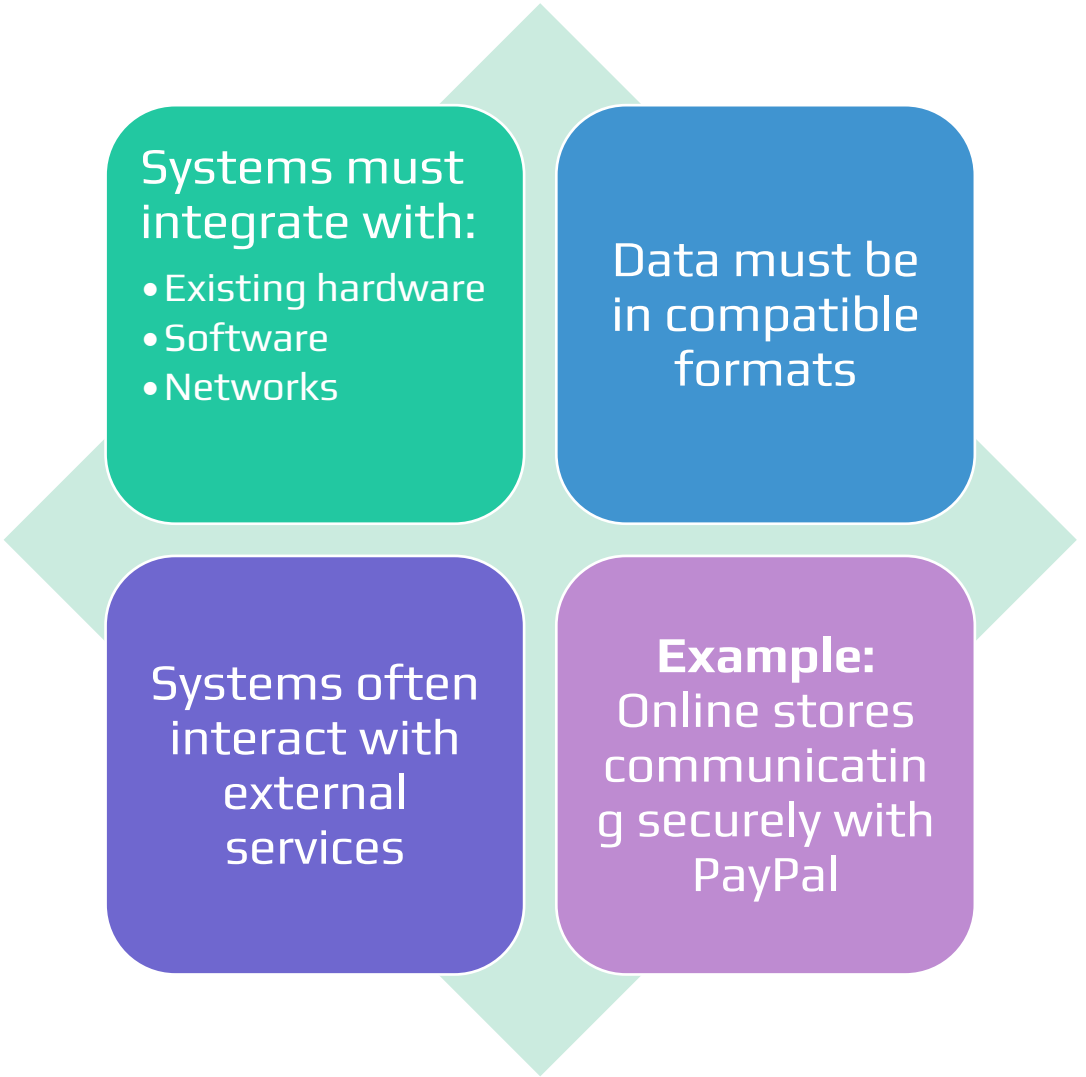
Systems must consider diverse user groups:

- Aboriginal and Torres Strait Islander Peoples
- Culturally and linguistically diverse communities
- People with disabilities

Important considerations:

- Avoiding data bias
- Accessible design (e.g. W3C standards)
- Respecting Indigenous Cultural and Intellectual Property (ICIP)

Technical Considerations



Legal Considerations

Copyright: The protection provided to the creators of original works and makers of sound recordings and films, that offers a legal framework for the control and reproduction or transmission of their literary, dramatic, artistic or musical works.

1

Systems must comply with laws such as:

- Privacy
- Copyright

2

Personal data must be protected using:

- Secure passwords
- Encryption

3

Designers must consider whether systems could:

- Enable illegal actions
- Be misused by users

Legal Considerations

Information privacy involves the establishment of rules governing the collection and handling of personal data such as credit information, and medical and government records. It is also known as ‘data protection



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Cybersecurity Principles

Cybersecurity: the protection of information technology elements, including hardware and software, data or network services.

Confidentiality: protect sensitive data

Integrity: ensure data is accurate and not altered

Availability: systems accessible when needed

These principles form the foundation of cybersecurity

Authentication, Authorisation and Accountability

Authentication is the process of verifying who a user is, while **authorisation** is the process of verifying what they have access to.



Authentication: verifies who the user is

Authorisation: controls what users can access

Accountability: records user actions

Implemented using:

- Passwords and biometrics
- Permissions and roles
- Logs and monitoring

Case Study – Shared E-Scooter System

